

Transient flow and noise characteristics of vortex-turbulence-noise interaction in centrifugal pump

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Abstract

Centrifugal pump operation noise is an avoidable common phenomenon in engineering practices, and its characteristics are closely influenced by the flow characteristics of the fluid. In this paper, a numerical simulation of centrifugal pump operation with different flow rates is carried out using the detached eddy simulation turbulence model and compared with the experimental results. Then the noise source and far-field radiated noise characteristics of the centrifugal pump are investigated using Lighthill's analogy. The results show that the noise of centrifugal pump is closely related to the flow characteristics of impeller, especially the generation and development of vortex. The relationship between turbulent kinetic energy (TKE) and vortex is complex, involving the generation, development and shedding of vortex. The influence of flow rate variation on vortex structure in impeller is studied by vortex transport equation. It is found that the relative vortex stretching (RVS) term is the main driving force of vortex distribution. The distribution of the Coriolis force (CORF) term is mainly concentrated in the inlet and outlet areas of the flow channel, which is related to the rotational motion of the impeller and the dynamic characteristic of the fluid. The power spectral density inside the impeller is mainly concentrated in the low frequency region, indicating that the low frequency noise component is dominant in energy. The directivity curve of far-field noise pressure level is influenced by hydrodynamic effect, impeller rotation direction and pump design structure, and will have a certain deviation. The spectral curve analysis of velocity, vorticity, pressure and sound pressure level shows that these physical quantities have strong correlation on the spectrum, in which the low frequency component is large, and the high frequency component is complex and volatile.

Keywords

Transient flow, noise, vortex-turbulence-noise interaction, detached eddy simulation

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Introduction

Centrifugal pumps can provide a large head supply and medium to high flow rate. Thus, they're widely implemented in power energy industry for both power generation and consumption.^{1–3} With rapid increased demand of centrifugal pumps in the consumption side, the stable operations of centrifugal pumps are essential in modern industry.⁴ Currently, centrifugal pumps must satisfy the various operation conditions to provide diverse power supplies or output demands. While, in different operating demands and conditions, centrifugal pumps are set in varied degrees of vibration, which in turn cause diverse operating noise. The noise further affects the stable operation of the pump. However, the operation of the pump requirements is also increasingly high to provide qualified power and outputs. The stable operation of the pump has become an important indicator for evaluating the performance of the pump. Therefore, the efficient, safe and stable operation of centrifugal pumps is a hot issue of great concern. It is of great significance to study the transient flow, noise, internal flow state and the interaction mechanism between them under different operating conditions of centrifugal pumps.

Fluid excitation, resulted from unsteady flow in centrifugal pumps is the main cause of vibration.⁵ The flow in centrifugal pumps is complex due to their design, making them highly prone to flow separation, vortex shedding, and secondary flows.^{6–8} The resulting non-constant pressure pulsations are a major source of fluid excitation.^{9–11} Generally, the pressure pulsation intensity can be used as an important reference for the vibration noise of centrifugal

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pumps with flow excitation.⁵ Radial clearance dimensions were analyzed to study the impeller-tongue interaction in centrifugal pumps. The flow fluctuations within the pump are influenced by variations in both radial clearances and flow rates.¹² Subsequently, many remarkable methods have been proposed to deal with the pressure pulsation suppression in centrifugal pumps. A deliberately designed position and deflection angle of the diverter blades can effectively suppress the water-excited vibration of centrifugal pumps.¹³ The non-uniform blade thickness distribution also reduces the pressure pulsation in the pump to a certain extent.¹⁴ The combination of impeller and split volute arrangement of staggered blade design can suppress pressure pulsation.¹⁵ The design for the trailing edge of the blades effectively reduces the vortex strength of the trailing edge of the blades, attenuates the effect of stator-rotor interaction, and further reduces the pressure pulsation amplitude.¹⁶ Changing the distribution of impeller blade thickness and blade angle to change the blade profile effectively reduces the hydraulic vibration.¹⁷ The above studies also show that the centrifugal pump impeller, as an important component, plays a decisive role in the performance and stable operation of the centrifugal pump.

Subsequently, flow-induced noise has begun to be studied extensively. Currently, experimental measurements of noise are expensive, and the results are also interfered by a variety of factors. Moreover, the separation of noise components is difficult,^{18,19} and therefore noise needs to be studied by numerical simulation. Computational Aerodynamic Acoustics (CAA) methods usually require significant computational resources, especially for far-field radiated noise. Lighthill's acoustic analogue theory-based methods are a potential alternative. In the computation of flow-induced noise, the effect of unsteady flow on it should be considered. Taking hydrofoils as an example, in the study of transient noise characteristics at accelerated flow conditions, the coherent mechanism of vorticity and turbulent kinetic energy was obtained.²⁰ The effect of unsteady flow in the different tip clearance size on near-field and far-field radiated noise.²¹ The study of sound source localization analysis in hydrofoils elucidated the vortex-acoustic interaction mechanism.²² Hydrodynamic noise of conduit propellers has also been investigated and the relevant role of vortices in the generation of rheological noise is discussed.²³ The effect of propeller tip vortices and underwater radiated noise was investigated and a correlation was found between the noise level and the tip vortex flow state.²⁴ The study of rotor speed fluctuations reveals that variations in speed can generate significant additional noise at specific frequencies.²⁵ Many researchers have also proposed measures to control the flow-induced noise. The flow-induced noise can be effectively reduced by small changes in the design structure.²⁶ Bionic structures are proposed to reduce the flow instability and further reduce the flow-induced noise.²⁷ The leading edge of the sawtooth structure has an effect in the mid to high-frequency range of the noise, reducing the far-field radiated noise by about 1.5 dB.²⁸ The double sawtooth structure is also effective in reducing the leading-edge noise.²⁹ The use of a separated micro-perforated fairing to reduce the turbulent interference noise generated by the leading edge of the airfoil was obtained, and the windward width and clearance distance can effectively reduce the noise.³⁰ The design

structure with grooves and wavy leading edges can also effectively reduce the interference noise of the airfoil.³¹ By studying the effect of sawtooth Angle on the three-dimensional flow field and noise reduction mechanism of airfoil, it is found that the airfoil with sawtooth wavelength to amplitude ratio of 0.3 shows the most effective noise reduction performance in the low frequency range.³² In summary, flow-induced noise is closely related to fluid dynamics.

Through the above literature, it is found that domestic and foreign scholars have conducted a large number of studies on the transient flow characteristics and vibration during the operation of pumps. However, the research on transient noise is more focused on hydrofoils, and there are fewer discussions on the intrinsic mechanism between the unsteady flow characteristics and induced noise in centrifugal pumps. This paper investigates the unsteady flow and transient noise characteristics of a centrifugal pump under varying operating conditions. The study examines the relationship between vorticity and turbulence within the pump impeller, and the connection between flow rate and vortex structures using vortex transport equations. Additionally, the paper analyzes the power spectral density and sound pressure level distribution of the impeller, along with the far-field radiated noise distribution in three directions at different flow rates. In this paper, the correlation mechanism between vortex-turbulence-noise is explored under different flow rates. Also, some of the correlation mechanisms are investigated. The work in this paper lays a foundation for an in-depth study of the transient noise mechanism of centrifugal pumps, as well as subsequent research on the design of centrifugal pumps with low vibration and low noise.

The rest of the paper is organized as follows. The experimental process, numerical simulation methods, including fluid simulation and noise simulation, and the relevant theories used in them are introduced respectively in Section Mathematical model and Geometric model and numerical method. In addition, in the results and discussions of Section Results and discussion, vortex of transient flow characteristics, near-field radiated noise analysis, far-field radiated noise, vortex-turbulence-noise interaction are also introduced analysis in subsequent sections.

Mathematical model

Turbulence model

The DES (detached eddy simulation) model combines elements of the RANS (Reynolds-averaged Navier-Stokes) and LES (large eddy simulation) formulations to obtain a hybrid formulation in which RANS is used within the connected and lightly separated boundary layer. In addition, LES is applied to the region of large-scale separation.

The DES model is based on the SST formulation. The advantage of this combination is that the accurate prediction of the turbulent boundary layer up to separation and in the region of mild separation is inherited from the SST model. In addition, the SST model supports the development of a regional DES formulation,³³ which is less sensitive to grid resolution limitations than the standard DES formulation proposed by Strelets.³⁴

The idea behind the DES model of Strelets³⁴ is to switch from the SST-RANS model to the LES model in regions

where the turbulence length L_t predicted by the RANS model is greater than the local grid spacing. In this case, the length scale used in the turbulent kinetic energy equation to calculate the dissipation rate is replaced by the local grid spacing Δ .

$$\varepsilon = \beta^* k \omega = k^{\frac{3}{2}} / L_t \rightarrow k^{\frac{3}{2}} / (C_{DES} \Delta) \text{ for } C_{DES} \Delta < L_t \quad (1)$$

$$\Delta = \max(\Delta_i); L_t = (\sqrt{k}) / \beta^* \omega \text{ for } C_{DES} \Delta < L_t \quad (2)$$

The practical reason for choosing the maximum edge length in the DES formulation is that the model should return the RANS formulation in the attached boundary layer. Thus, the maximum edge length is the safest estimate to ensure demand.

The DES correction for Strelets can be expressed as a multiplier of the destructive term in the k -equation.

$$\epsilon = \beta^* k \omega \rightarrow \beta^* k \omega \cdot F_{DES} \text{ for } F_{DES} = \max\left(\frac{L_t}{C_{DES} \Delta}, 1\right) \quad (3)$$

C_{DES} is equal to 0.61 because the limiter should only be valid in the model region. The numerical formulas are also converted between upwind bias and center difference formats in the RANS and DES regions, respectively.

Noise source model

The variational formulation of the Lighthill's analogy was first derived by Oberai et al.^{35,36} The strong variational statement associated with equation (4) can be written as,

$$\int_{\Omega} \frac{\omega^2}{c^2} \psi \delta \psi d\Omega + \int_{\Omega} \frac{\partial^2 \psi}{\partial x_i \partial x_i} \delta \psi d\Omega = \int_{\Omega} \frac{1}{i\omega} \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j} \delta \psi d\Omega \quad (4)$$

where $\delta \psi$ is a test function and Ω is a part of the computational domain (non-moving and non-deforming). Partial integration of the spatial derivatives using Green's theorem yields the weak variational form of the spatial derivatives,

$$\begin{aligned} \int_{\Omega} \frac{\omega^2}{c^2} \psi \delta \psi d\Omega - \int_{\Omega} \frac{\partial \psi}{\partial x_i} \frac{\partial \delta \psi}{\partial x_i} d\Omega &= \int_{\Omega} \frac{i}{\omega} \frac{\partial \delta \psi}{\partial x_i} \frac{\partial T_{ij}}{\partial x_j} d\Omega \\ &- \int_{\Gamma} \frac{\delta \psi}{i\omega} \frac{\partial}{\partial x_i} (c^2 \rho \delta_{ij} + T_{ij}) n_i d\Gamma \end{aligned} \quad (5)$$

The expression of T_{ij} is,

$$\tilde{T}_{ij} = \tilde{\rho} \tilde{v}_i \tilde{v}_j + \delta_{ij} (\tilde{p} - a_0^2 \tilde{\rho}) + \tilde{\tau}_{ij} \quad (6)$$

Substituting the right-hand side of equation (6) for T_{ij} in the surface integral, and utilizing the momentum equation, equation (5) becomes

$$\begin{aligned} & - \int_{\Omega} \frac{\omega^2}{\rho_0 c^2} \psi \delta \psi d\Omega - \int_{\Omega} \frac{1}{\rho_0} \frac{\partial \psi}{\partial x_i} \frac{\partial \delta \psi}{\partial x_i} d\Omega \\ &= \int_{\Omega} \frac{i}{\rho_0 \omega} \frac{\partial \delta \psi}{\partial x_i} \frac{\partial T_{ij}}{\partial x_j} d\Omega - \int_{\Gamma} \frac{1}{\rho_0} \mathcal{F} (\tilde{\rho} \tilde{v}_i n_i) d\Gamma \end{aligned} \quad (7)$$

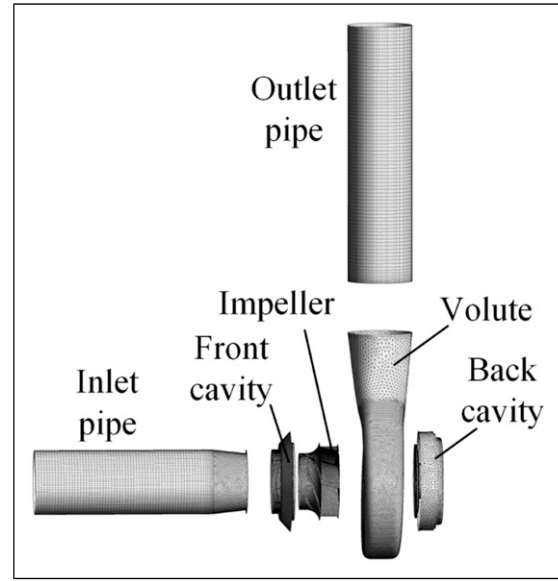


Figure 1. Grid of centrifugal pump.

If the surface Γ is fixed or vibrating in its own plane, then $-n_i \frac{\partial}{\partial t} (\rho v_i)$ decreases to zero and the right-hand side of equation (7) vanishes. This corresponds to the natural boundary conditions of the weak variational problem. Therefore, for surfaces fixed or vibrating in their plane, no surface source term is needed.

Geometric model and numerical method

Geometry structure of centrifugal pump

The computational model and grid distribution of the centrifugal pump shows in Figure 1, ANSYS 2020R2 is used for modeling, meshing, and subsequent flow field simulation. The computational model includes the inlet pipe, the front cavity, the impeller, the volute, the back cavity and the outlet pipe. Among them, the structured mesh is selected in the inlet and outlet pipe, the unstructured mesh is selected in the front cavity, impeller, back cavity and volute, and the boundary layer mesh is drawn for the wall part to ensure the accuracy of the computation results. At the same time, due to the consideration of the pump in the actual operation of the impeller wear ring gap leakage flow and the flow of the front and back cavity, to ensure that the computation results are more in line with the flow state of the pump. Table 1 shows the structural parameter of the centrifugal pump.

Flow simulation

The DES is an intermediate technique between LES and the RANS method. The method captures the turbulence structure in the large-scale separation region while computing the lightly separated boundary layer with the RANS method. The method has a higher resolution than the RANS equations and is therefore used for numerical calculations.

The boundary conditions were set up in a controlled experiment. The pressure inlet was set at the entrance, the flow outlet was set at the exit, and the wall was set as No Slip Wall. The maximum number of iterations was verified

Table 1. The structural parameter of the centrifugal pump.

Description	Parameter	Units	Value
Design head	H	m	20
Design flow rate	Q_d	m ³ /h	160
Rotational speed	n	r/min	2960
Specific speed	n_s	-	240
Number of blades	z	-	6
Impeller inlet diameter	D_1	mm	110
Impeller outlet diameter	D_2	mm	148
Impeller outlet width	b_2	mm	36
Blade inlet angle	β_1	°	25
Blade outlet angle	β_2	°	15
Blade thickness	s	mm	4
Volute base circle diameter	D_3	mm	196
Volute inlet width	b_3	mm	50
Volute outlet diameter	D_4	mm	125

by simulation and finally set to 10. To verify the settings of the experiment, the other parameters were set exactly in accordance with the experiment. Among them, for the discrete processing of pressure gradient terms, the second-order upwind style is used to ensure accuracy.

In order to discuss the flow variations within the impeller in detail, five monitoring points, I01-I05, were set up along the flow center from the impeller inlet to the outlet, as shown in Figure 2.

Validation of simulation with mesh and time-step

In order to verify the reliability of numerical simulation, the centrifugal pump computation model is verified for grid-independence. By comparing the operating head and efficiency of the centrifugal pump to verify that the change in the number of grids does not affect the simulation results, show in Figure 3. Among them, the number of grids varies from 3.2 million to 5.7 million. From Figure 3, the simulation results of 4 sets of grids have less variation. Therefore, the 3rd set of grid is selected with a grid number of 4.6 million to make the calculation balanced between independence, speed and accuracy.

Secondly, the effect of time step on the accuracy of the numerical simulation results was also investigated. Three time steps were selected as 5.631×10^{-5} s, 1.126×10^{-4} s, and 2.252×10^{-4} s, it corresponds to $T_i/360$, $T_i/180$, $T_i/90$, T_i represents the rotation period of the impeller. Figure 4 demonstrates the change process of pressure fluctuation at the monitoring points I01 and I02 of the impeller channel of the centrifugal pump, and the pressure change of the centrifugal pump at these three time steps is highly consistent, indicating that the influence on the three time steps can be negligible. The accuracy of this calculation is therefore verified.

Sound simulation

The sound source in the acoustic simulation is determined by the computation of the non-constant flow field, using a computational process that combines CFD with acoustic boundary elements, as shown in Figure 5.

The acoustic computational domain consists of three components: the fluid domain, the structural domain and the air domain of the pump, as shown in Figure 6. The modeling, meshing and simulation of the acoustic model were done using Actran 2020. Among them, the air domain is set as a flat surface at the bottom of the pump, which simulates the ground under the pump. Specifically, for acoustic predictions around the noise source, a convex hull with a distance of 0.2 m was established centered on the pump body to enclose it within an air domain.

Results and discussion

Comparison with experimental results

Centrifugal pump experimental bench is located in Wenzhou City, Zhejiang Province, Kaiquan Group Co., Ltd type test center, the experimental bench mainly consists of inlet pipe, test pump section, motor, outlet pipe, photoelectric speed and torque meter, electromagnetic flowmeter, inlet and outlet pressure sensors, etc. shown in Figure 7. In the centrifugal pump experiment, the first control flow stabilized after the experimental data stabilization, data acquisition. The test sketch of the experimental bench is shown in Figure 7(b), which demonstrates the overall and main equipment of the experimental bench.

In order to demonstrate the consistency of the centrifugal pump simulation results with the experimental results, Figure 8 includes the operation results of five different flow points, namely, $0.4Q_d$, $0.6Q_d$, $0.8Q_d$, $1.0Q_d$, and $1.2Q_d$, whose numerical computations are set up to be consistent with the experimental working condition settings. The specific head $\psi = gH/U_2^2$, where U_2 is the circumferential velocity at impeller outlet. From Figure 8, the numerical computation results are in good agreement with the experiments. Under the small flow condition (when $Q = 0.8Q_d$), the difference between the computation head and the experiment is 0.009, and the difference between the computed efficiency and the experiment is 0.54%. When the flow rate increases to $1.2Q_d$, the difference between the simulated results increases and the distribution is 0.029 and 1.44%. From Figure 8, the numerical simulation results and experiments on the whole change trend has been, verifying the accuracy of the numerical calculation method.

Vortex of transient flow characteristics

Figure 9 shows the vorticity and streamline distributions in the midplane of the impeller for flow rates $0.4Q_d$, $0.6Q_d$, $0.8Q_d$, $1.0Q_d$, and $1.2Q_d$. The vortex is closely related to the vorticity, and the center of the vortex tends to correspond to a relatively large vorticity intensity. And the vortex intensity has a high correlation with the flow rate. When the flow rate is $0.4Q_d$, the flow inside the impeller is complicated and the vortex intensity is larger. The larger vortex intensity and vortex appear in both the pressure and suction surfaces of the impeller within the flow channel. Among them, the range of vortices appearing at the trailing edge of the

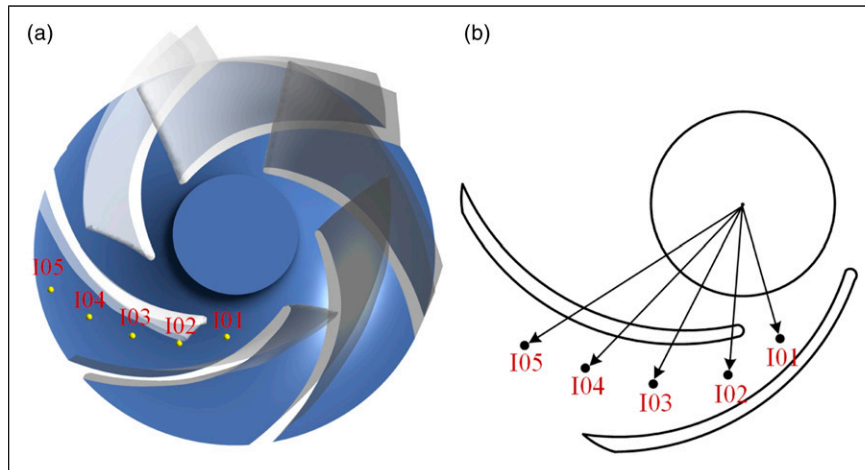


Figure 2. Setting of monitoring points in the middle flow path of the impeller (I01-I05 from the impeller inlet to the outlet, respectively).

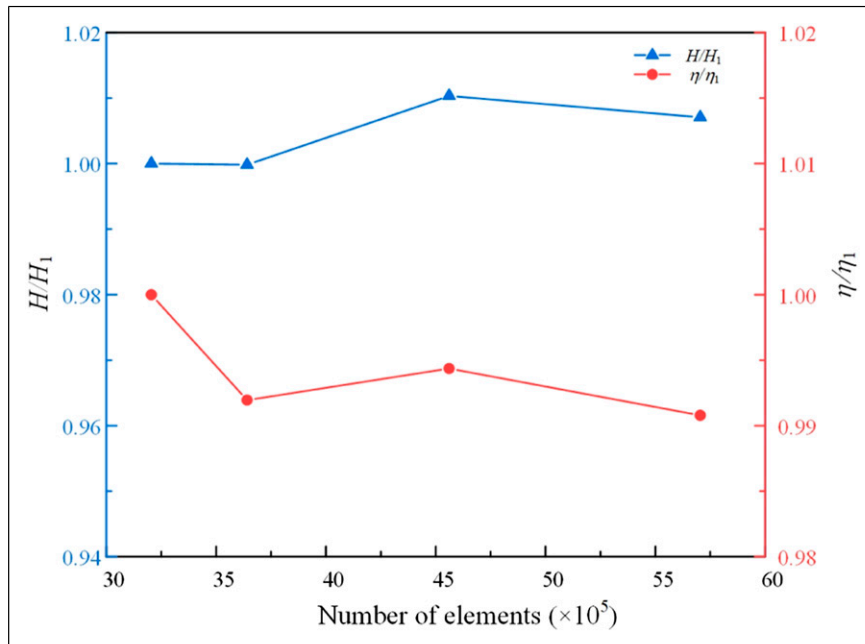


Figure 3. Mesh density independence test.

suction side of the blade is larger, which is due to the backflow that occurs as a result of the increased flow cross section of the flow channel and the weakened control of the fluid by the blade.

When the flow rate is $0.6Q_d$, the vortex appearing at the trailing edge of the blade pressure surface disappears, but the vortex appearing at the trailing edge of the suction surface remains. At the same time, the vorticity is reduced. It shows that the increase in flow rate weakens the unstable vortex mass in the pump. When $Q = 0.8Q_d$, the vortices at the trailing edge of the blade suction surface also weakened. A region of increased vorticity also begins to appear near the pressure surface of the blades, which is due to the shedding of unstable vortex caused by the flow separation that occurs at the leading edge of the blades. When $Q = 1.0Q_d$, the vorticity in the impeller further decreases, primarily concentrating on the reflux area observed at the trailing edge of the suction surface of

the blade and the unstable flow occurring along the streamline direction at the pressure surface of the blade. When $Q = 1.2Q_d$, this is greater than the design flow rate of the operating conditions, the pump large-scale vorticity distribution area is further reduced. The high intensity vorticity distribution is also closer to the blade region. This indicates that the flow rate of the fluid between the impeller channels is also enhanced by the increase in flow rate. Influenced by the blade boundary, the flow separation area induced by the fluid at the leading edge of the blade increases, and a high-intensity vorticity distribution is formed. At the same time by the high flow rate of high-speed fluid wrapped around the impeller outlet.

Turbulent kinetic energy (TKE) is another way to describe the stability of the flow. Figure 10 illustrates the relationship between vortices and TKE. It can be seen that the relationship between vortices and TKE is

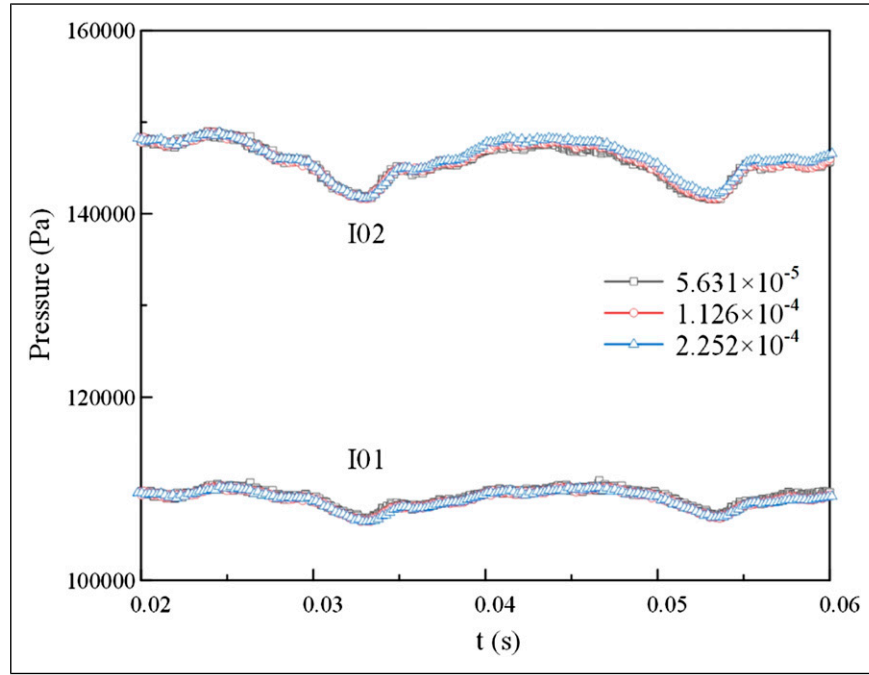


Figure 4. Time step independence test.

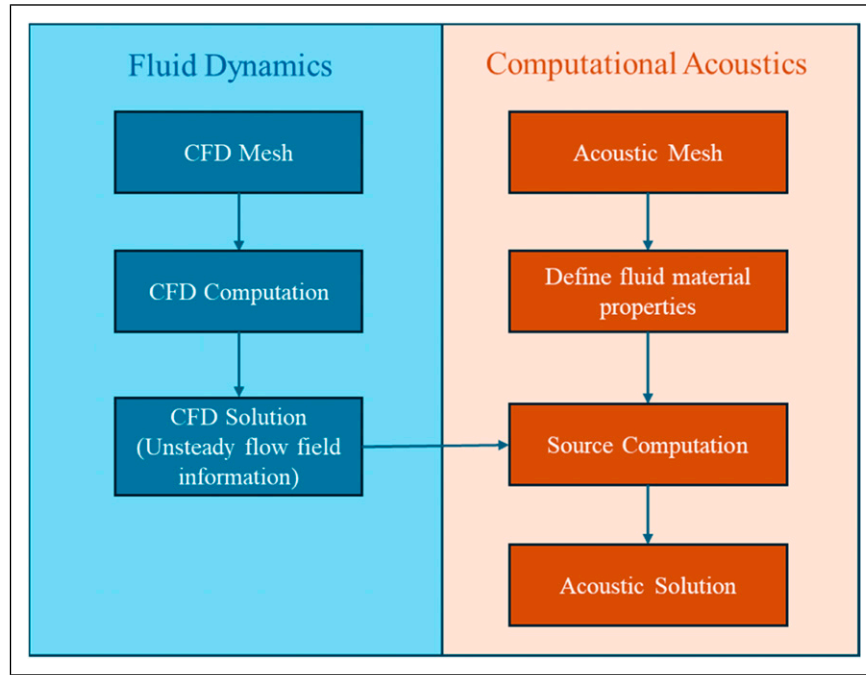


Figure 5. Computational processes for acoustics and fluid dynamics.

complex. There is some correlation between the TKE corresponding to the vortices in the impeller channel, especially at $Q = 0.4Q_d$. However, the correlation between vortices and TKE gradually decreases when the flow rate increases. Comparison of Figure 9 shows that the value of vortices near the blade surface is large, but the corresponding TKE value is small.

In order to discuss in detail the distribution of the vortex structure of the impeller channel within the impeller and to explore the internal mechanism of the flow-induced noise, the vortex distribution was investigated by utilizing the vorticity transport equation.^{37–39}

$$\frac{D\vec{\Omega}_r}{Dt} = (\vec{\Omega}_r \cdot \nabla)\vec{W} - \vec{\Omega}_r(\nabla \cdot \vec{W}) - 2\nabla \times (\omega \times \vec{W}) + \frac{\nabla \rho_m \times \nabla p}{\rho_m^2} + \nu \nabla^2 \vec{\Omega}_r \quad (8)$$

where $\vec{\Omega}_r$ means the relative vorticity, $\vec{W}$ means the relative velocity, and ν represents the kinematic viscosity.

Each term on the right-hand side of the above equation represents a different vector. These are the relative vortex stretching term (RVS) which represents the stretching and bending effects of the vortex line caused by the velocity gradient, the expansion term (RVD) which represents the

effect of the volume change of the fluid on the size of the vortex, the baroclinic torque term (BART) which represents the effect of asymmetry between the pressure gradient and the density gradient, the viscous diffusion term (VISD) which represents the viscous diffusion effect of the vorticity, and the rotational effect produced by the Coriolis force (CORF).

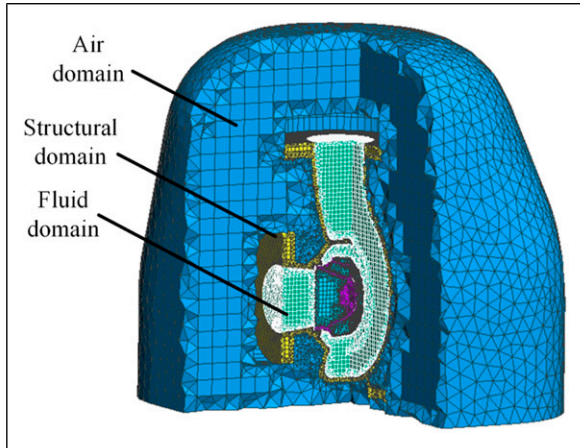


Figure 6. Acoustic computational domain settings (including fluid domain, structural domain and air domain).

Figure 11 illustrates the distribution of RVS within the impeller flow channel for different flow rates. The distribution of the RVS term fills most of the runner area within the impeller, especially near the blades, which correlates with the vortex intensity distribution. It shows that the RVS term is the main driver of the vortex distribution within the impeller. The RVS distribution is more fragmented at $Q = 0.4Q_d$. A tendency for the RVS term to move into the flow channel occurs near the middle of the suction surface of the blade. The comparison with Figure 9 reveals a wide range of vortices present, exerting an influence on the distribution of vorticity. This is further compounded by the gradual widening of the flow channel, the progressive expansion of the overflow area, and a diminishing ability of the blade to effectively control fluid dynamics, resulting in decreased flow stability. As the flow rate increases, the RVS distribution in the impeller channel gradually weakens, and the RVS distribution near the blades gradually increases. When $Q = 1.0Q_d$, the RVS distribution in the impeller channel decreases, and the vortex distribution with the surface at the design flow condition is lower. Subsequently, at $Q = 1.2Q_d$, the RVS near the blade surface increases, and the flow separation produced by the fluid influenced by the blade increases, which is consistent with the conclusion obtained in Figure 9.

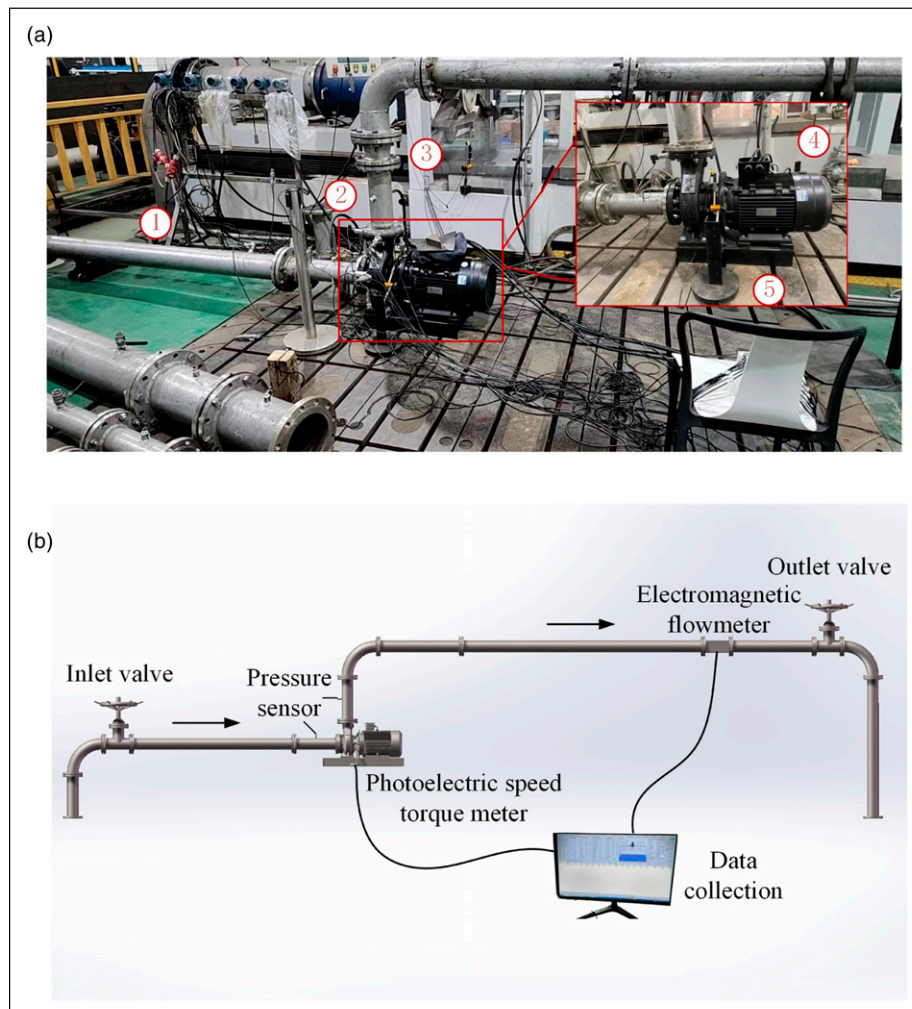


Figure 7. Centrifugal pump test bench. (1- Water inlet pipe; 2- Test pump section; 3- Outlet pipe; 4- Motor; 5- Photoelectric speed torque meter).

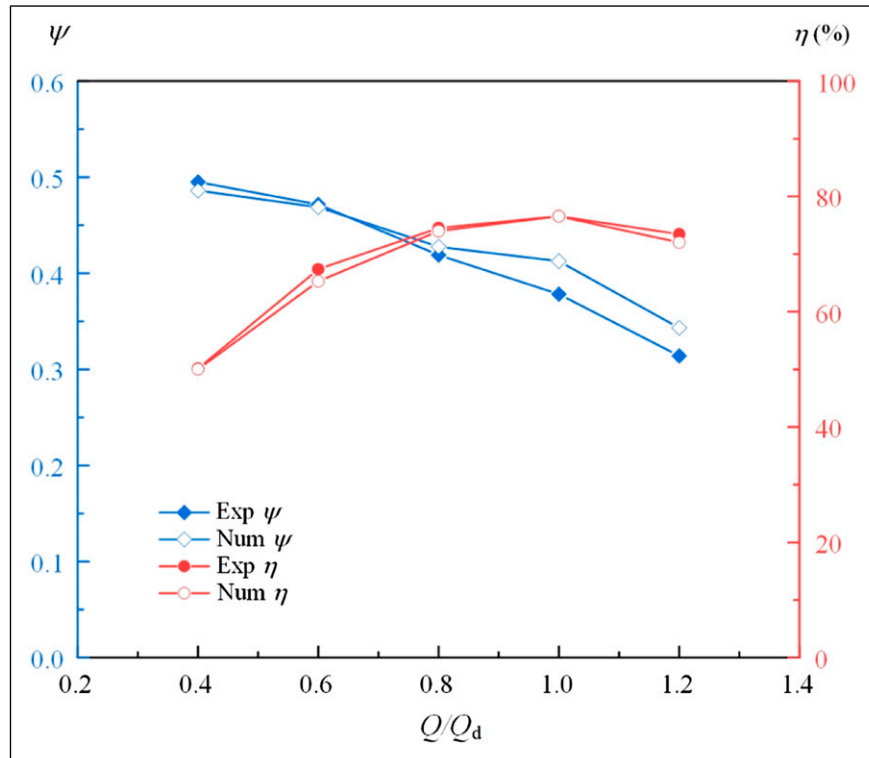


Figure 8. Comparison of centrifugal pump experiment and numerical simulation.

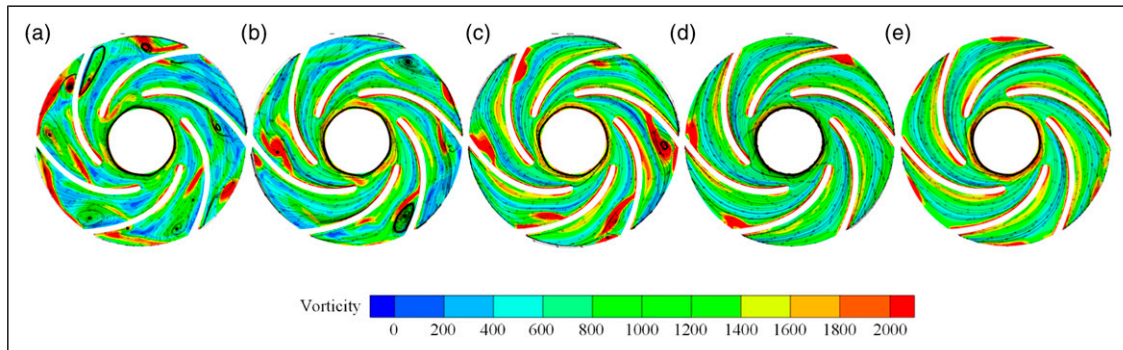


Figure 9. Vorticity and streamline diagram of impeller at $0.4Q_d$, $0.6Q_d$, $0.8Q_d$, $1.0Q_d$, and $1.2Q_d$.

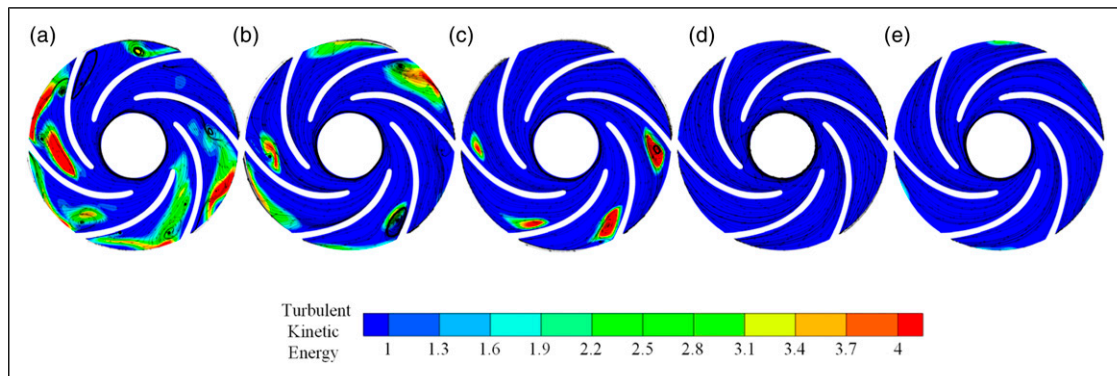


Figure 10. TKE and streamline diagram of impeller at $0.4Q_d$, $0.6Q_d$, $0.8Q_d$, $1.0Q_d$, and $1.2Q_d$.

Figure 12 illustrates the distribution of CORF term within the impeller channels for different flow rates. The distribution of CORF is more sporadic. At $Q = 0.4Q_d$, the distribution of CORF appears in the pressure and suction

surfaces at the leading and trailing edges of the blade, showing a strong asymmetry. It shows that the flow is complicated at this time, and the rotational effect influences the generation and development of vortices in the impeller,

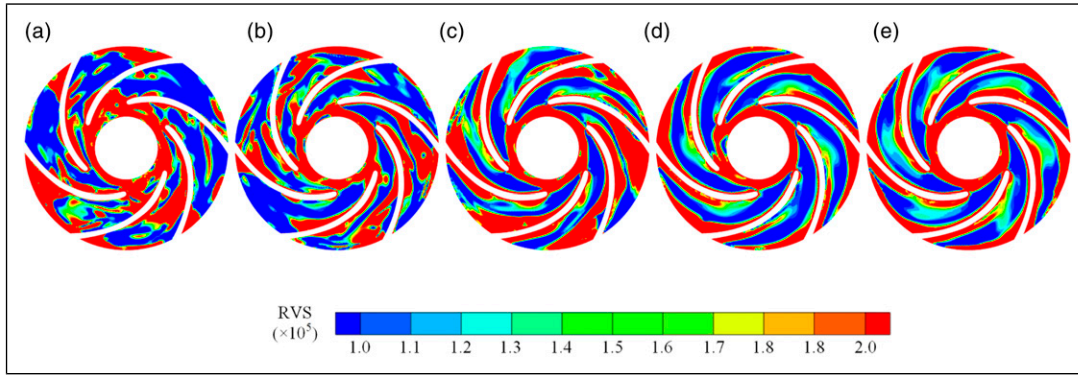


Figure 11. RVS term distribution of impeller at $0.4Q_d$, $0.6Q_d$, $0.8Q_d$, $1.0Q_d$, and $1.2Q_d$.

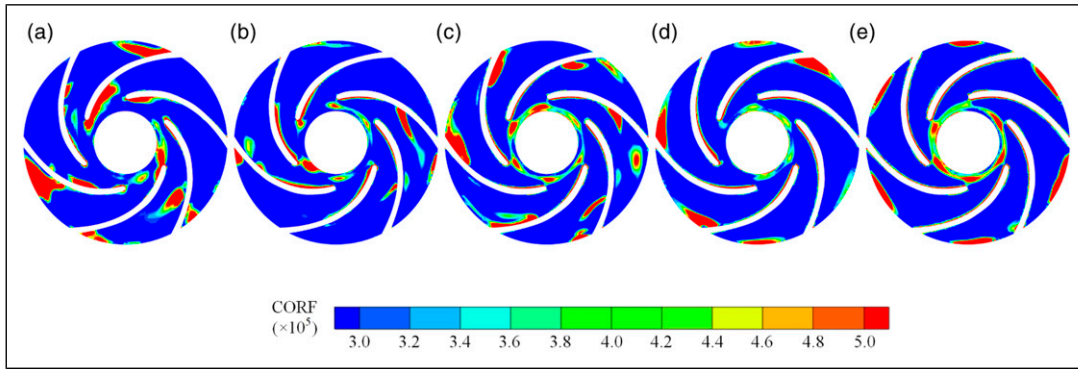


Figure 12. CORF term distribution of impeller at $0.4Q_d$, $0.6Q_d$, $0.8Q_d$, $1.0Q_d$, and $1.2Q_d$.

especially in the reflux area appearing on the suction surface side of the blade. The distribution of CORF gradually stabilizes as the flow increases, and the distribution at the leading edge of the blade is weakened, but a symmetrical region of high values appears at the impeller inlet plane. This indicates a gradual stabilization of the flow. At both $Q = 0.8Q_d$ and $1.0Q_d$, a larger region of high CORF distribution appears at the trailing edge of the blade suction plane. Comparing with Figure 9, this indicates that the generation of a large area of vortices at the trailing edge of the blade suction surface is also influenced by the CORF. When $Q = 1.2Q_d$, the CORF distribution in the impeller channel is further weakened and appears only in the impeller inlet and outlet flow channels. This indicates the reduced influence of CORF on the flow state within the impeller channel at high flow conditions.

The above analysis indicates that the interactions between TKE, RVS, and CORF are complex and strongly correlated with flow changes. For example, increasing the flow will reduce the correlation between RVS and TKE, but at the same time increase the influence of CORF. In particular, in the low flow region, the influence of these three parameters is more concentrated in the impeller passage due to the instability of the flow. However, as the flow increases, the distribution of these three parameters gradually approaches the blade area. This is due to the influence of flow separation and vortex generation and development at low flow conditions. Furthermore, it can be seen that at design flow of $1.0Q_d$ and high flow of

$1.2Q_d$, the high value distribution of TKE is basically dispersed, indicating that the energy dissipation of the flow is lower. The distribution of RVS and CORF is still relatively high, especially with an increase in the blade inlet. This further indicates that the distribution of TKE in a centrifugal pump is highly correlated with the flow. The distribution of RVS and CORF, in addition to the flow, is also related to the operating conditions and blade parameter design of the pump.

In order to study the effect of different flow rates in the impeller on the pressure fluctuation in the impeller channel, five pressure monitoring points were set up in the impeller channel: I01, I02, I03, I04, and I05. The setup of the impeller monitoring points is shown in Figure 2.

For a discrete-time signal $x[n]$, the discrete Fourier transform (DFT) is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-\frac{2\pi}{N}kn} \quad (9)$$

Where N is the number of samples in the signal, and k is the frequency index.

The power spectral density $P[k]$ can be calculated as follows:

$$P[k] = \frac{1}{N} |X[k]|^2 \quad (10)$$

Figure 13 demonstrates the power spectral density (PSD) distribution of I01-I05 at different flow rates. PSD is a function used to describe the power distribution of a signal

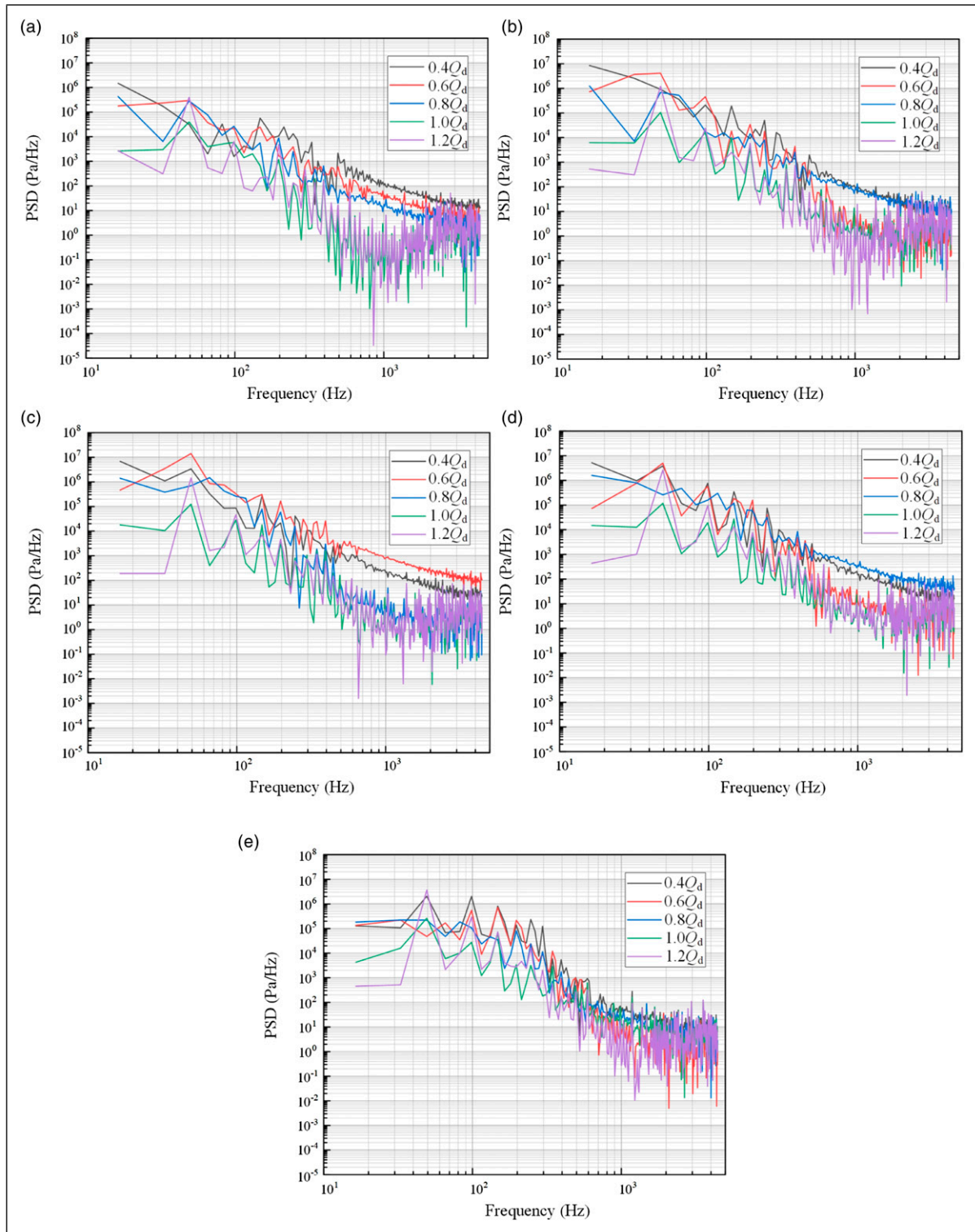


Figure 13. Power spectral density of monitoring points at different Q : (a) I01, (b) I02, (c) I03, (d) I04, and (e) I05.

in the frequency domain. It represents the amount of power in a unit bandwidth of the signal and is often used to analyze random signals. It is a physical quantity that characterizes the relationship between the power energy of a signal and its frequency. The power spectral density is the expected value of the square of the Fourier transform of the signal, and it provides information on the distribution of the signal's power at different frequencies. It shows the maximum power spectral density is mainly concentrated at low frequencies. The maximum power spectral density occurs at

$f = 49$ Hz, which corresponds to the shaft frequency of the centrifugal pump. Subsequently, additional peaks emerge in the frequency doubling of the shaft frequency. This observation suggests that there is a higher concentration of power energy at both the shaft frequency and its harmonics. The power spectral density at high frequencies is smaller. From the full frequency range, the power spectral density distribution decreases first with increasing flow rate, and the power spectral density is lowest at $Q = 1.0Q_d$ design flow rate, and then increases at high flow rate.

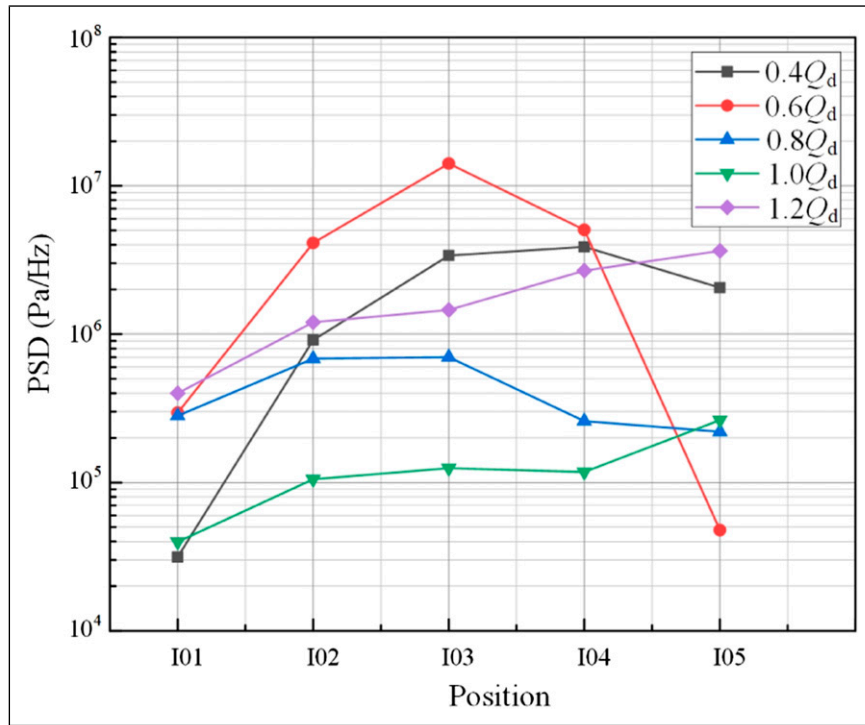


Figure 14. Maximum PSD of five monitoring points at different Q .

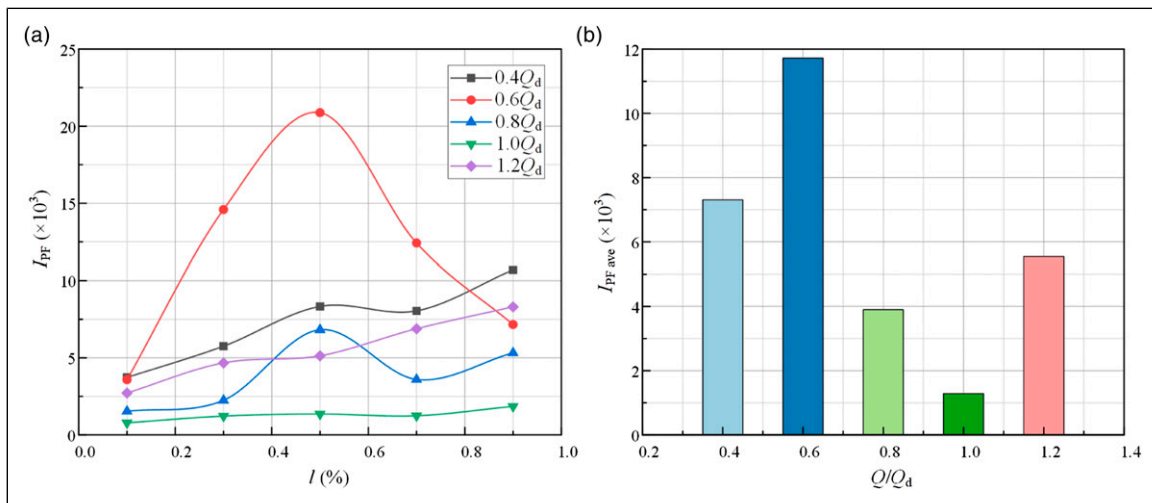


Figure 15. Pressure fluctuation intensity distribution. (a) Pressure fluctuation intensity on the impeller channel from leading edge to trailing edge. (b) Average pressure fluctuation intensity distribution at different flows).

In order to discuss in detail the distribution of PSD at low frequencies, Figure 14 illustrates the distribution of maximum PSD at five monitoring points at different flow rates. The maximum power spectral density shows a distribution trend of increasing and then decreasing, with a significant increase in the value of power spectral density at point I03, followed by a rapid decrease. It indicates that the flow complexity is higher at point I03. After the fluid enters the impeller, the flow velocity is enhanced by the influence of impeller rotation and blade control. However, due to the gradual increase of the overflow cross section, the blade's ability to control the fluid is weakened, and the fluid acceleration process has not been completely completed, so the flow stability in the middle part of the impeller channel is lower and the flow state is more complex. Comparison of

Figure 10 impeller plane TKE distribution can also be seen in the middle part of the impeller channel appeared in the impeller TKE distribution of high value areas. And at $Q = 1.0Q_d$, there is a significant decrease in the PSD distribution. The main reason for this phenomenon is that at this time the flow rate is the design flow rate and the flow stability is higher. However, at $Q = 1.2Q_d$, the flow rate exceeds the design flow rate of the pump, and the distribution of the maximum value of the PSD is lower than the distribution at small flow rate. In the process of I01-I05, the maximum value of PSD is gradually increasing, which indicates that the flow complexity in the impeller in this condition is low and there is no unsteady turbulence in the impeller channel, which is consistent with the results obtained from the TKE distribution.

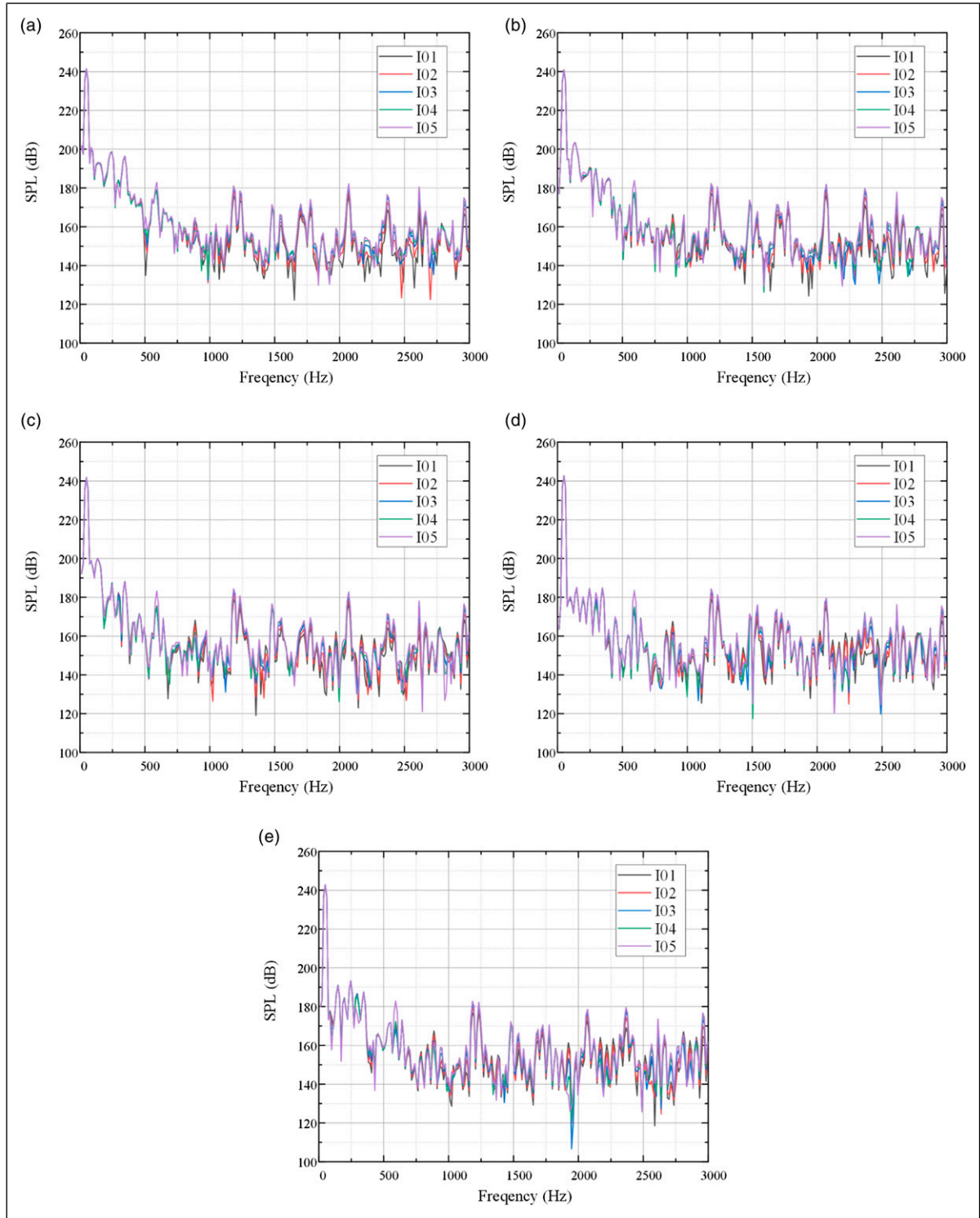


Figure 16. SPL spectrum curve of monitoring points at different Q : (a) $0.4Q_d$, (b) $0.6Q_d$, (c) $0.8Q_d$, (d) $1.0Q_d$, and (e) $1.2Q_d$.

The root mean square method is used to define the pressure fluctuation intensity (I_{PF}) as shown in equations (11) and (12):

$$\bar{p} = \frac{1}{M} \sum_{i=1}^M p_i \quad (11)$$

$$I_{PF} = \sqrt{\frac{1}{M} \sum_{i=1}^M (p_i - \bar{p})^2} \quad (12)$$

where M is the number of sampling points, p_i is the pressure at each time step and $\bar{p}$ is the arithmetic mean the root of the pressure.

The distribution of pressure fluctuation intensity in the impeller channel is illustrated in Figure 15, where 15(a) shows the distribution at different flow rates from the leading edge to the trailing edge of the impeller channel. On the whole, this is similar to the maximum PSD distribution in Figure 14, which shows that the fluctuation intensity is lower at the impeller inlet, and the pressure fluctuation is gradually enhanced with the accelerated

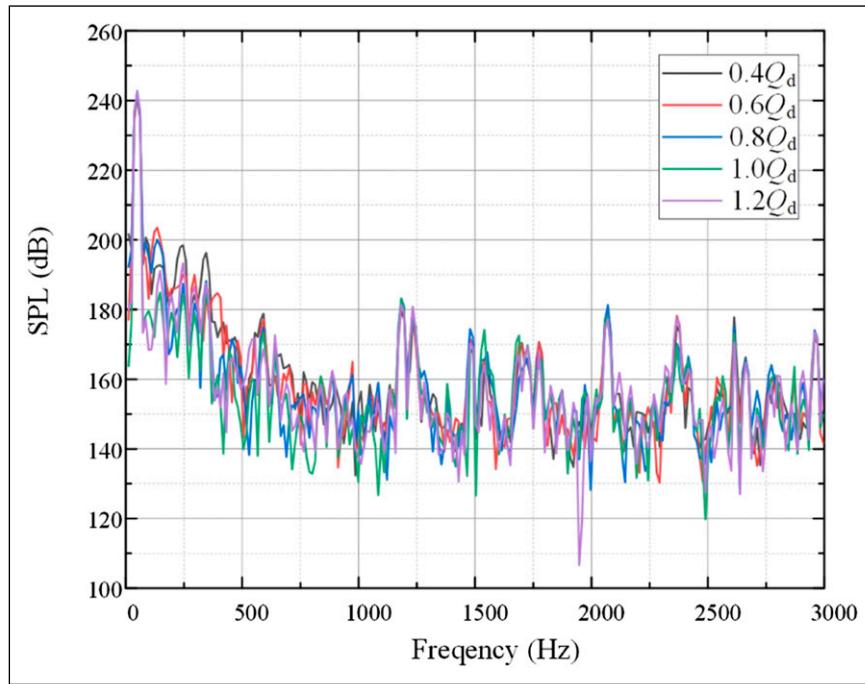


Figure 17. Maximum SPL of I03 monitoring point for different Q .

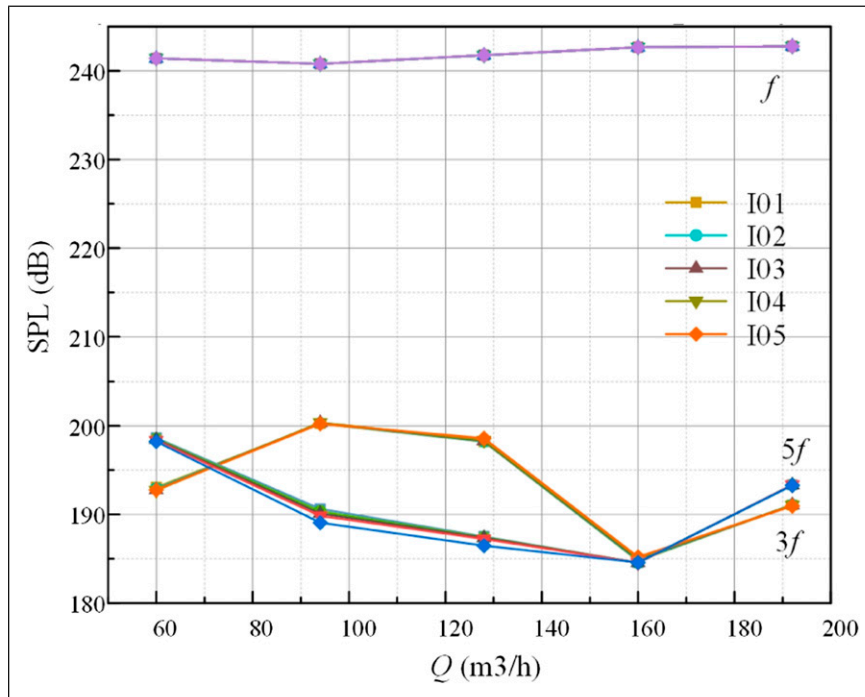


Figure 18. Maximum SPL of five monitoring points for f , $3f$, $5f$.

movement of the fluid in the impeller channel. This also shows that the maximum PSD distribution represents the overall pressure fluctuation to a certain extent. Meanwhile, at flow rates less than the design flow rate $Q = 1.0Q_d$, the pressure fluctuations in the middle of the impeller channel at $l = 0.5$, that is point I03, are all increased to a certain extent, which indicates that the flow stability in the middle of the impeller channel is weaker. The intensity of pressure fluctuations in the flow path

decreases at design flow rate and at high flow rate $Q = 1.2Q_d$. The vortex intensity and streamline distribution described in the previous Figure 9 are well explained. In terms of the flow inside the impeller, at flow rates lower than the design flow rate, the unsteady flow starts to develop in the flow channel from the vicinity of the blades due to the presence of vorticity and backflow in the flow channel. While in the design flow rate and large flow rate, by the influence of the flow rate increase, the vortex

intensity distribution in the flow channel is close to the development of the blade. This different flow state in the flow channel leads to a different distribution of pressure fluctuation intensity at different flow rates. This is also reflected in the distribution of the mean pressure fluctuation intensity, as shown in Figure 15(b). The average pressure fluctuation intensity shows a tendency of first increasing, then decreasing and then increasing with the increase of flow rate. This is consistent with the distribution of vortex intensity in the impeller channel with flow rate trend, especially at $Q = 0.6Q_d$, the vortex and backflow region increases (as shown in Figure 9(b)). The reasons for the exceptionally high I_{PF} of $0.6Q_d$ also include that at $0.4Q_d$, due to the lower flow rate, the overall velocity distribution is also lower. When the flow rate increases, the velocity gradually rises from the inlet to the outlet of the impeller. When the flow rate reaches $0.6Q_d$, the velocity at the inlet of the impeller is relatively high, while the velocity at the outlet remains within a lower range. That is, the large velocity gradient causes the instability of the impeller flow field. It can also be observed from Figure 15(a) that the intensity of pressure fluctuation at $0.6Q_d$ rises rapidly from the inlet, reaches the peak in the middle of the impeller passage and then drops rapidly. With the further increase of the flow rate

and the gradual reduction of the velocity gradient, the flow field gradually tends to be stable.

Near-field radiated noise analysis

In order to investigate the effect of different flow rates on the near-field noise inside the pump impeller, five monitoring points in the impeller channel I01-I05 were selected. As shown in Figure 2, the monitoring points are equally distributed along the flow path between them.

Figure 16 shows the near-field sound pressure level (SPL) distribution at monitoring points I01 to I05. The maximum SPL values are mainly concentrated at low frequency ($f = 49$ Hz) and the SPL values are small at high frequency. This is because $f = 49$ Hz is the shaft frequency of the pump, indicating that the noise in the pump is mainly caused by rotation. This indicates that for different flow rates, the SPL values at the near-field monitoring points are mainly concentrated at low frequencies. Comparing the SPL distribution under different monitoring points, it is found that the SPL distribution of the monitoring points in the flow channel has a high correlation at low frequencies, but starts to differ at high frequencies. The SPL values gradually increase as they approach the impeller outlet, e.g., in Figure 16(a), at $f = 500$ Hz at $Q = 0.4Q_d$, it is found that the noise distributions of I01-I05 are gradually elevated. This indicates the strong velocity and pressure pulsations induced in the impeller outlet region as the fluid accelerates with the rotation of the impeller. From Figure 13, it is evident that the pressure pulsation at I05 exhibits multiple prominent peaks, while its PSD demonstrates a slower rate of decrease with increasing frequency compared to I01-I04. Comparison of the SPL distributions at different flow rates shows that different distributions begin to appear at the octave frequency of f . As a whole, two distinct SPL peaks appear at $Q = 0.4Q_d$, and the number of peaks decreases as the flow rate increases. This shows that the low frequency noise is a little more complex at small flow rates. When $Q = 1.0Q_d$, the distribution of SPL octave frequencies is much lower. At $Q = 1.2Q_d$, the SPL distribution increases. This also shows that the increased flow rate due to increased flow further affects the SPL distribution.

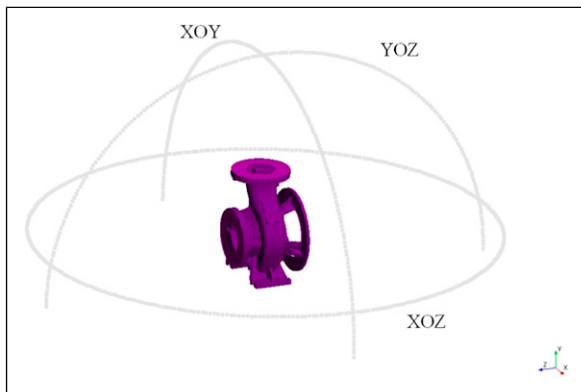


Figure 19. The monitoring points of acoustic field at XOY, YOZ and XOZ plane.

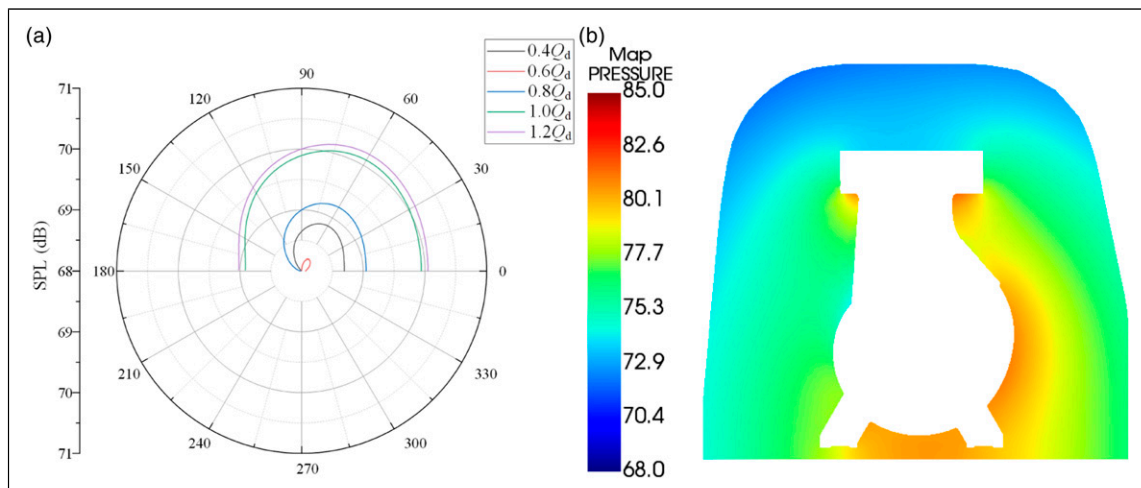


Figure 20. Directivity of SPL for XOY plane at $f = 49$ Hz. (a) for five different Q ; b. SPL directional distribution).

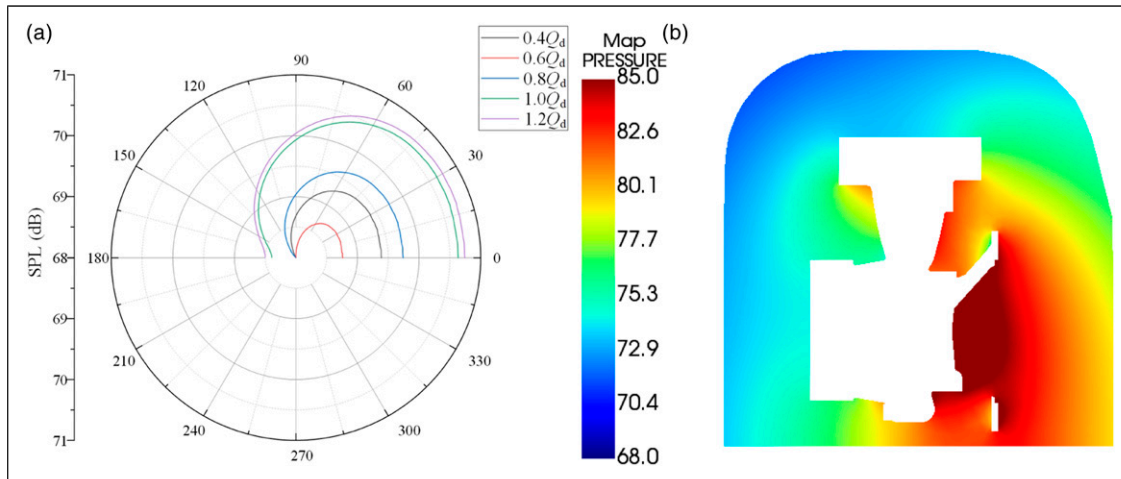


Figure 21. Directivity of SPL for YOZ plane at $f = 49$ Hz. (a) for five different Q ; (b) SPL directional distribution).

The pressure fluctuation at point I03 is more drastic as described above, and in order to study the relationship between pressure fluctuation and SPL distribution, the SPL distribution at monitoring point I03 is analyzed. Figure 17 demonstrates the SPL distribution at monitoring point I03 at different flow rates, and there is one major amplitude and three minor amplitudes at low frequencies for the SPL distribution at different flow rates. At the three minor amplitudes, it can be seen that the SPL distribution at low flows is slightly higher than the SPL distribution at high flows. As the frequency increases, the SPL decreases and starts to show different distributions at different flow rates. This shows that the difference in SPL distribution at different flow rates is mainly concentrated at the high frequency noise.

In order to discuss the SPL distribution under different frequencies, the SPL values under the main frequency f and octave frequencies $3f$ and $5f$ for monitoring points I01-I05 at different flow rates were selected as shown in Figure 18. The distribution of SPL under main frequency f has less variation, while under $3f$ and $5f$, the SPL distribution has more variation. As a whole, the SPL under $5f$ is lower than the SPL distribution under $3f$. Its variation relationship with flow rate also shows that at $Q = 1.0Q_d$ design flow rate, the $3f$ and $5f$ SPL decreases by an average of 13.38 dB and 2.65 dB. Also, it can be seen from the $5f$ SPL distribution that the I01-I05 SPL distributions start to show differences as well. This shows that the noise distribution is mainly concentrated in the low frequency in the impeller channel.

Far-field radiated noise analysis

In order to investigate the effect of different flow rates of the centrifugal pump on the far-field noise characteristics, three propagation directions were established for analysis. In order to quantify the directionality of the pump's radiated noise in the far-field, 180, 180 and 360 monitoring points centered on the impeller were established, and the SPL of each monitoring point was analyzed. The monitoring points were uniformly distributed in a circle with a radius of 1 m, as shown in Figure 19. Among them, the XOY and YOZ

planes are semicircles, which are set due to the fact that the bottom of the pump is installed on the ground.

Figure 20(a) shows the SPL directivity curves at 49 Hz for 180 monitoring points in the XOY plane, which are distributed in an irregular semicircle. The SPL is relatively large at 60-75 degrees of the pump and relatively small at 150-180°. This is mainly because of the asymmetric shape of the volute, the fluid enters the volute through the impeller, and the rotational effect has a certain impact on the pump. With the change of flow rate, the SPL distribution first decreases and then increases, and the SPL is minimized at $Q = 0.6Q_d$. This is due to the fact that the vortex intensity is greater at $Q = 0.4Q_d$, and the vortex exists in both the blade pressure and suction surfaces (as shown in Figure 9(a)). Whereas, at $Q = 0.6Q_d$, the increase in flow rate increases the flow stability in the impeller, the vortex intensity decreases and the vortex near the blade pressure surface disappears (as shown in Figure 9(b)). The decrease in vortex intensity and the increase in flow stability lead to a decrease in the SPL distribution at $Q = 0.6Q_d$. With further increase in flow rate, the flow stability in the pump is further enhanced. The differentiation of the flow field within the impeller at different flow rates decreases, and the change of the flow field at high flow rates is limited by the pump structure, and the change is more reflected in the distribution of velocity. That is, the flow rate increases, the pump overflow area remains unchanged, and the flow rate increases. This also leads to the SPL distribution in the high flow area ($Q = 1.0Q_d$ and $Q = 1.2Q_d$) when the sound propagation law is very similar.

In order to study the sound propagation distribution of the pump, the SPL distribution in the XOY plane outside the pump for $f = 49$ Hz is plotted at $Q = 0.8Q_d$ as shown in Figure 20(b). The SPL distribution outside the pump is in the high-pressure region on the right side and below the pump, while on the left side it is at a lower sound pressure level. This is due to the structural asymmetric design of the pump and the rotational direction of the pump impeller. The impeller throws the fluid out and into the volute, impinging on the flow path of the volute and flowing out towards the outlet under the guiding effect of the volute. At the same time, the presence of the floor prevents sound propagation

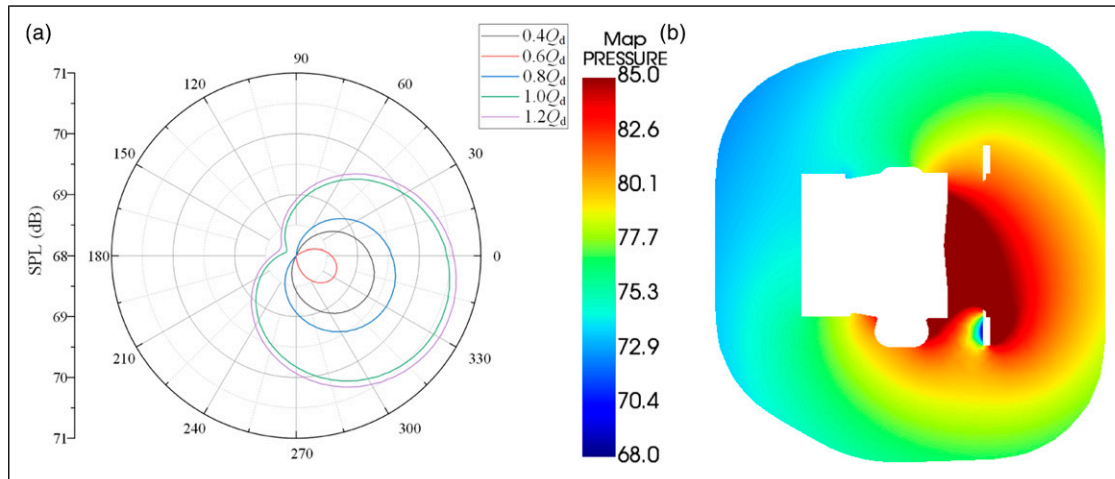


Figure 22. Directivity of SPL for XOZ plane at $f = 49$ Hz. (a) for five different Q ; (b) SPL directional distribution).

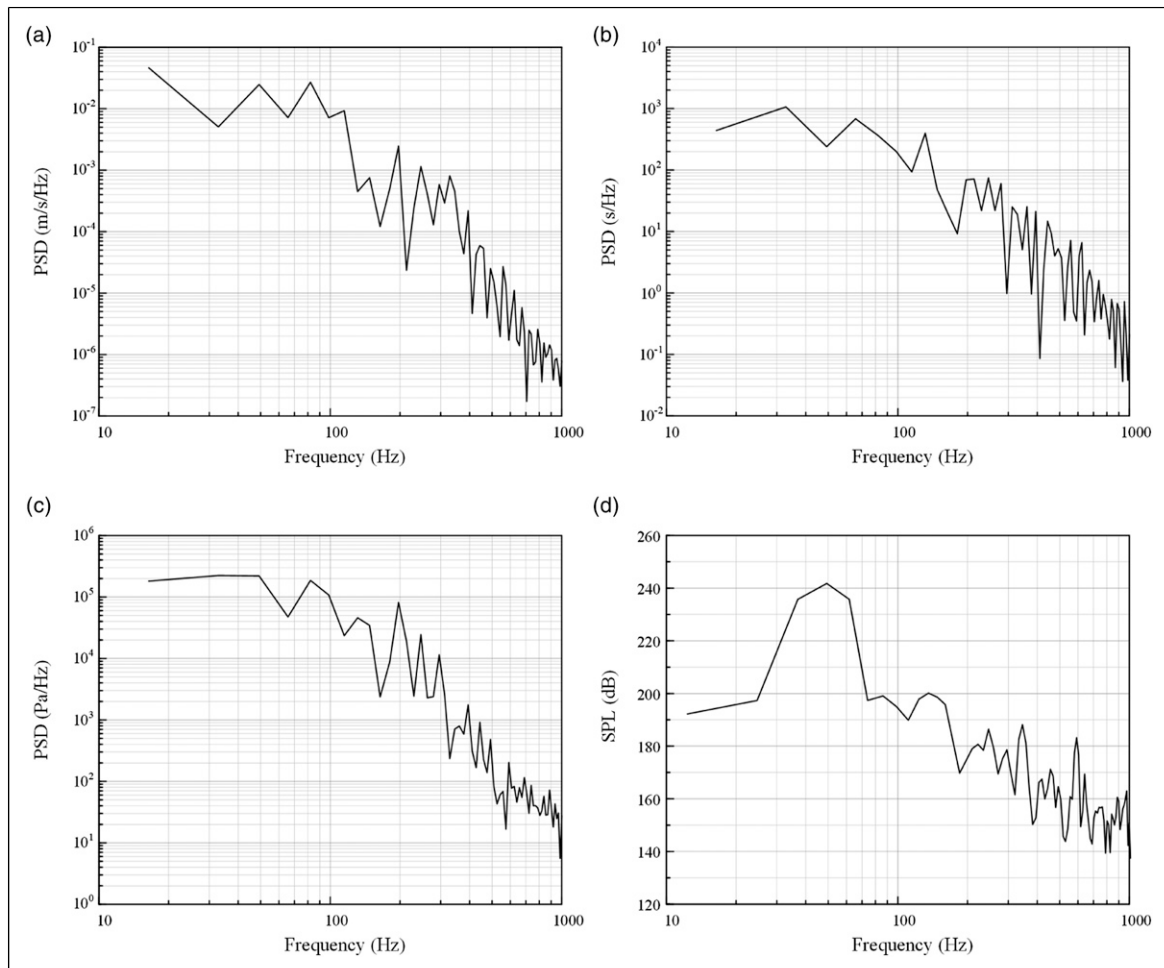


Figure 23. PSD distribution of velocity, vorticity and pressure, SPL distribution of near field (I05 monitoring point at $Q = 0.8Q_d$).

from traveling downward, further exacerbating the sound pressure level below the pump.

Figure 21(a) shows the SPL directivity curves at 49 Hz for 180 monitoring points in the YOZ plane, which are distributed in an irregular semicircle. The SPL is relatively large at 0-30 degrees of the pump and relatively small at 165-180°. This is mainly due to the impact on the pump caused by the fluid entering the pump. Its change rule with the flow and the distribution of Figure 20(a) in the

distribution of similar, SPL distribution first decreased and then increased, with the difference is that the distribution of the YOZ plane by the flow of the vortex in the influence of the smaller, more by the impact of the fluid.

The SPL distribution in the YOZ plane outside the pump for $f = 49$ Hz at $Q = 0.8Q_d$ is shown in Figure 21(b). The SPL distribution outside the pump is in the high-pressure region on the right side of the pump, while on the left side it is at a lower SPL. This is due to the impact of the pump by

the fluid, and after the reflection of the ground, the SPL distribution is small on the left side and large on the right side, that is, the high SPL is mainly propagated to the rear of the pump.

Figure 22(a) shows the SPL directivity curves at 49 Hz for 360 monitoring points in the XOZ plane, which are distributed in two semicircles symmetrically distributed along the 150-degree and 330-degree lines. The SPL is relatively large at 330 degrees of the pump and smallest at 150°. This is mainly due to the fact that after the fluid enters the pump, the SPL is deflected in one direction by the combined effect of rotation and impact action.

The SPL distribution in the XOZ plane outside the pump for $f = 49$ Hz at $Q = 0.8Q_d$ is shown in Figure 22(b). The SPL distribution outside the pump is in the high-pressure region at the lower right side of the pump. This is caused by the rotation and impact of the pump by the fluid, which corresponds to the SPL distribution at 330° in Figure 22(a). At the same time by the pump blocking the influence of the upper left side is in the lower SPL, corresponding to Figure 22(a) 150 degrees of the SPL distribution. SPL distribution is the upper left side of the small, the lower right side of the distribution of large, that is, the high SPL is mainly propagated to the pump diagonal rear.

Vortex-turbulence-noise interaction

In order to investigate the correlation of the spectral characteristics of velocity, vortex, pressure and noise, the velocity field, vorticity, pressure field and near-field SPL at the monitoring point I05 at $Q = 0.8Q_d$ were analyzed shown in Figure 23. The spectral curves of velocity, vorticity, pressure and near-field SPL show similar trends. This indicates that the spectral properties of these physical quantities represent consistent correlations. At low frequencies, the value fluctuates over a wide range with a peak frequency of 49 Hz, while at high frequencies, the value has a smaller distribution and a complex frequency component with greater fluctuation.

Conclusions

Noise is an important index for the evaluation of stable operation of centrifugal pumps, especially in terms of flow-induced noise under different flow conditions. In this paper, DES model and Lighthill's analogy are used to numerically compute the transient flow and noise of centrifugal pumps. Also, the effects of different flow rates on the energy, flow and noise characteristics of centrifugal pumps are investigated. The main conclusions are as follows.

In terms of vortex structure, pump impeller vortex and vorticity are closely related to the vortex center often corresponds to a relatively large vortex intensity, and vortex intensity and flow rate has a high correlation. It is found by vortex transport equation the RVS term is the main driver of the vortex distribution. The distribution of RVS term is concentrated near the blades at $Q = 1.0Q_d$, it shows that the flow is not uniform. The distribution of the CORF term is more fragmented, and there is a strong asymmetry in the CORF distribution. It gradually stabilizes at high flow rates, concentrating in the inlet and outlet regions of each impeller

channel. In summary, when $Q = 1.0Q_d$, the distribution of RVS is concentrated on the blade surface, and the distribution of CORF is more in the inlet and outlet area of the impeller.

In terms of flow characteristics in the impeller, the maximum power spectral density in the impeller is mainly concentrated at the low frequency, this indicates that the low-frequency component is dominant in the flow energy. With the increase of flow, the PSD value decreases. At the monitoring point (such as I03) in the middle of the impeller channel, the pressure fluctuation intensity is larger and the correlation with the flow rate is higher, indicating that the flow complexity here is greater. Therefore, the central part of the impeller channel is the main generating position of unsteady flow.

Acoustically, inside the impeller, the distribution of sound pressure levels shows significant peaks in the low frequency region (e.g. 49 Hz), which are due to flow instability in the impeller or the breaking of eddy currents. The asymmetric design structure of the pump body leads to the directional deviation of the far field noise. Far-field noise is also affected by fluid shock and the direction of impeller rotation, which cause the noise to be more pronounced in a particular direction of the pump. The spectral curves of velocity, vorticity, pressure and sound pressure levels show a consistent correlation. This means that the energy of the noise is mainly concentrated in a specific frequency and spatial region. Therefore, the optimization design of centrifugal pump noise reduction can be studied from two aspects: reducing flow instability and improving pump structure design.

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